



C A P S T O N E P R O J E C T

Fake News Detection with Explanation

An end-to-end ML pipeline combining classical baselines, BiLSTM, DistilBERT, RAG retrieval, and FLAN-T5 explanation generation — with a Streamlit UI.

5-Member Team · ISOT + LIAR Datasets · 2025

AGENDA

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Problem Statement

Why fake news detection matters and our scope

02

Dataset & Preprocessing

ISOT + LIAR datasets, cleaning, semantic chunking

03

Model Architecture

Baselines → BiLSTM → DistilBERT transformer

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RAG + Explanation Layer

FAISS retrieval + FLAN-T5 / Gemini generation

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Results & Comparison

Accuracy, F1, macro-F1 across all models

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System Pipeline & UI

End-to-end flow and Streamlit interface

07

Conclusion

Key findings, limitations, and future work



Problem Statement

6 in 10

Americans encounter
fake news weekly

\$78B

Economic damage from
misinformation annually

~60%

Social media users
can't identify fake news

What This Project Solves

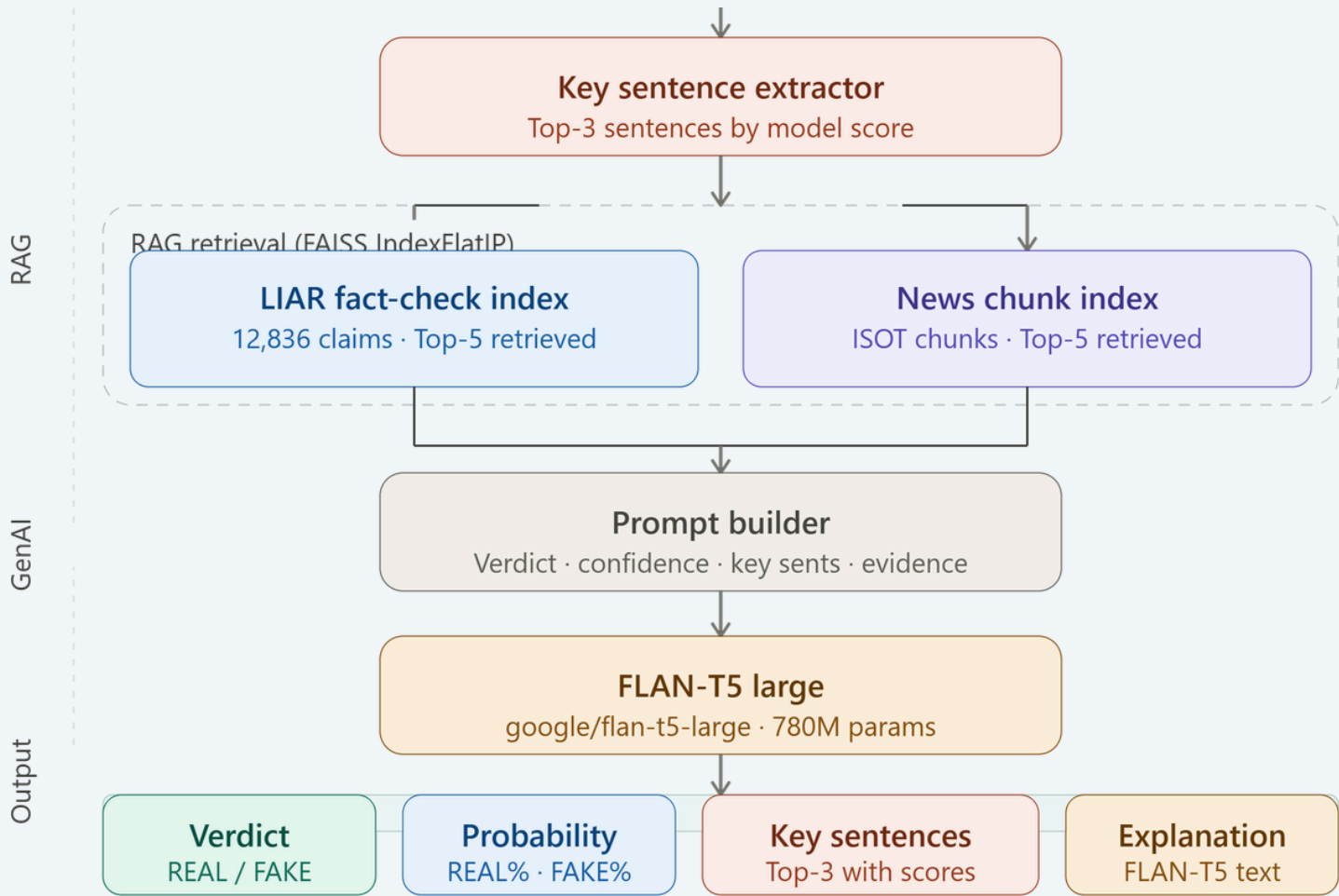
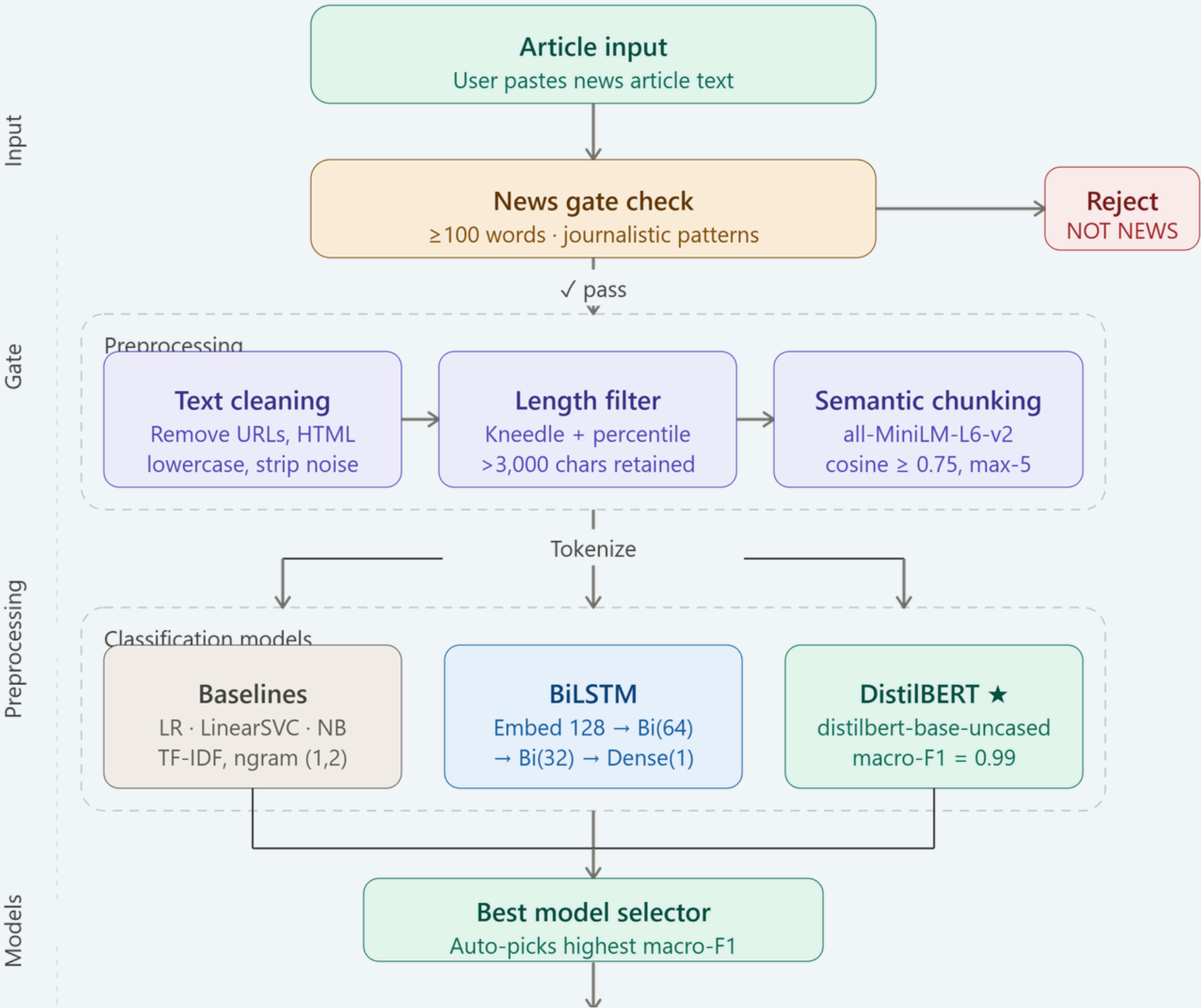
- ▶ Detect whether a news article is REAL or FAKE using trained ML models
- ▶ Provide a probability confidence score — not just a binary label
- ▶ Highlight the key sentences that most influenced the verdict
- ▶ Retrieve semantically similar fact-checked claims from the LIAR dataset via RAG
- ▶ Generate a human-readable explanation using FLAN-T5 LLM

Datasets & Preprocessing

ISOT Dataset	
Primary Training Corpus	
44,919	total articles
21,417	REAL articles (label = 1)
23,502	FAKE articles (label = 0)
55% / 45%	class balance — no imbalance
> 3,000 chars	threshold after preprocessing
70 / 15 / 15	train / val / test split

LIAR Dataset	
RAG Fact-Check Knowledge Base	
12,836	political fact-check statements
6 labels	true / mostly-true / half-true...
FAISS	IndexFlatIP for semantic search
all-MiniLM-L6-v2	sentence embedding model
Top-5 retrieval	per query during inference
ISOT + LIAR	combined for full pipeline

Solution flow



Data Preprocessing Pipeline

1 Raw Text Ingestion

True.csv + Fake.csv → unified schema with label (1=REAL, 0=FAKE). Both raw_text and clean_text preserved in DataFrame.

2 Text Cleaning

Remove URLs, HTML tags, special characters, numbers. Lowercase. Extra whitespace stripped. Source patterns (Reuters, AP) removed.

3 Length Filtering

Kneedle + percentile method identifies elbow. Articles < 3,000 chars removed — retains top 80% of data, reduces noise and training complexity.

4 Semantic Chunking

Sentences split via NLTK. Cosine similarity threshold (0.75) with all-MiniLM-L6-v2 groups coherent sentences into max-5 chunks for RAG corpus.

5 Tokenization

Keras Tokenizer (50k vocab, MAX_LEN=512) for BiLSTM. DistilBERT AutoTokenizer (max_len=256) for transformer. Both applied after cleaning.

Model Architecture

Baseline Models

TF-IDF + Classical ML

- ▶ Logistic Regression (max_iter=2000)
- ▶ Linear SVC
- ▶ Multinomial Naive Bayes
- ▶ TF-IDF: 20k features, ngram (1,2)
- ▶ Purpose: benchmark reference

BiLSTM

Deep Sequential Model

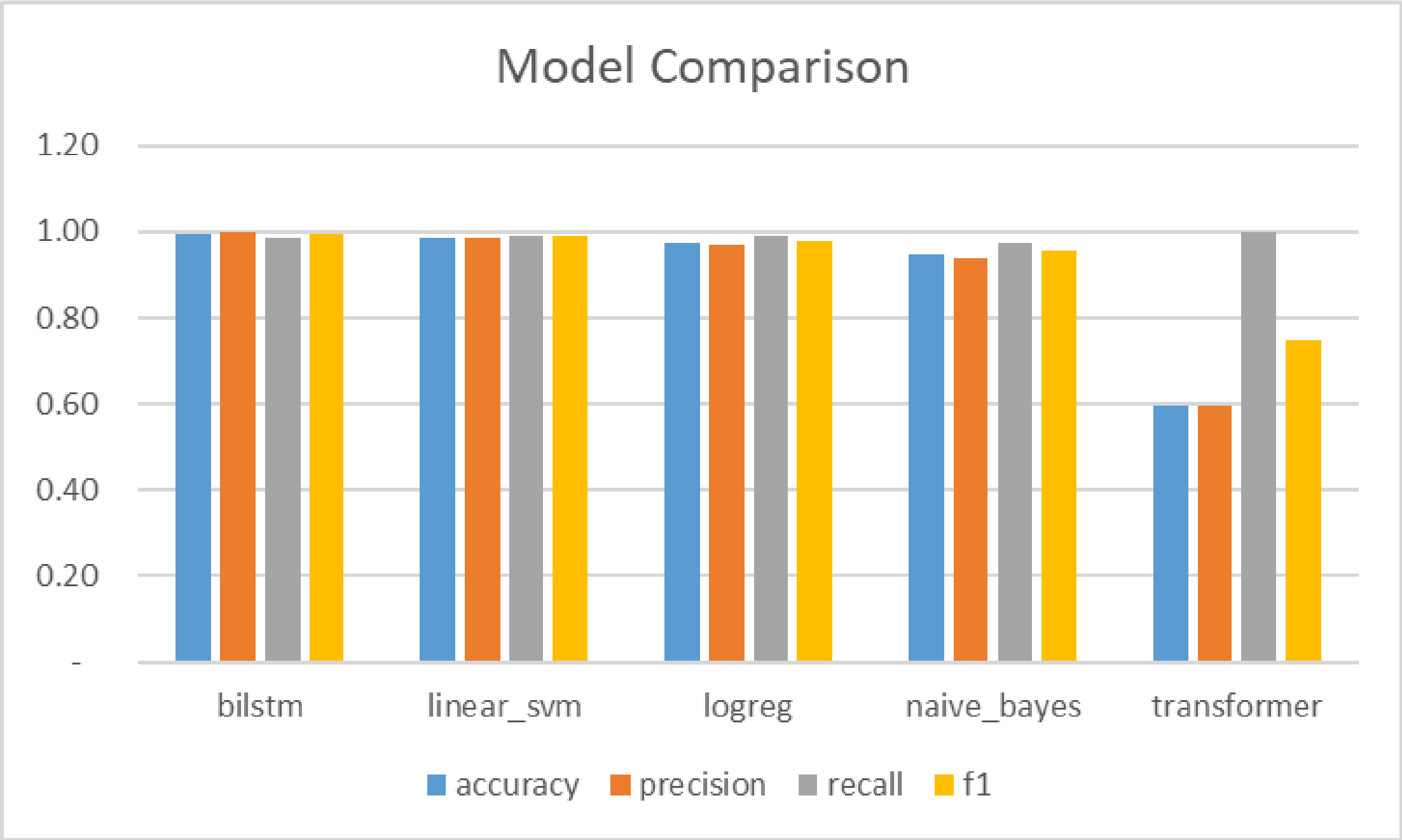
- ▶ Embedding: 50k vocab, 128-dim
- ▶ BiLSTM(64) → Dropout(0.3)
- ▶ BiLSTM(32) → Dropout(0.3)
- ▶ Dense(64, ReLU) → Dense(1, Sigmoid)
- ▶ MAX_LEN=512, EarlyStopping(patience=2)

Transformer - DistilBERT

Transformer (Fine-tuned)

- ▶ distilbert-base-uncased
- ▶ AutoModelForSequenceClassification
- ▶ max_len=256, batch=8, epochs=2
- ▶ learning rate 0.01, Weight Decay 0.01

Model Results & Comparison



	accuracy	precision	recall	f1
bilstm	0.993051	1.000000	0.988318	0.994125
linear_svm	0.987491	0.986079	0.992991	0.989523
logreg	0.974288	0.968000	0.989486	0.978625
naive_bayes	0.948575	0.940315	0.975467	0.957569
transformer	0.594858	0.594858	1.000000	0.745969

- Bilstm is better model in terms of accuracy.
- Due to model complexity of transfoimer it is trained only on 2 ephocs
- Upon increasing epochs there is possibility of Improvement

Note: Scores shown are on the validation set. Final model selected by macro-F1 automatically.

RAG Retrieval + Explanation Layer

Article Input

Raw text from user

Key Sentences

Top-3 by model score

FAISS Retrieval

LIAR + News FAISS indices

Prompt Builder

Verdict + evidence assembled

FLAN-T5 Output

Human explanation generated

LIAR RAG Retrieval

- 12,836 political fact-check statements embedded
- Embedding: all-MiniLM-L6-v2 (384-dim)
- FAISS IndexFlatIP for inner product search
- Top-5 claims retrieved per key sentence
- Labels: true / mostly-true / half-true / barely-true / false / pants-on-fire
- Provides factual grounding for explanation

FLAN-T5 / Gemini Explanation Generation

- Model: google/flan-t5-large (780M params) / Gemini
- Task: instruction-following seq2seq
- Prompt includes: verdict, confidence, key sentences
- Retrieved LIAR + news chunk evidence in context
- Output: 5-8 sentence explanation in natural language
- num_beams=4, max_new_tokens=300, min_new_tokens=80

End-to-End System Pipeline

User Input

Paste article
≥100 words

News Gate

Word count +
language check

Classification

Best model
predict_proba

Key Sentences

Top-3 scored
sentences

RAG Retrieval

LIAR + News
FAISS search

FLAN-T5

Explanation
generation

UI Output

Verdict + prob
+ explanation



Verdict Badge

Large REAL/FAKE badge, colour-coded green or red



Probability

Dual bars: REAL% and FAKE% from model confidence



Key Evidence Sentences

Top-3 flagged sentences with per-sentence scores



LIAR Evidence Table

Top-5 retrieved claims, colour-coded by truthfulness



AI Explanation

FLAN-T5 paragraph explaining verdict in plain English



News Gate Warning

Rejects non-news / short inputs before model call

Technology Stack

Data & ML

- ▶ pandas, numpy
- ▶ scikit-learn (LR, SVM, NB)
- ▶ TF-IDF (20k features, ngram 1-2)
- ▶ tensorflow / keras (BiLSTM)
- ▶ NLTK (sentence tokenization)

Deep Learning

- ▶ PyTorch
- ▶ HuggingFace Transformers
- ▶ distilbert-base-uncased
- ▶ google/flan-t5-large
- ▶ HuggingFace Trainer API

RAG & Retrieval

- ▶ sentence-transformers
- ▶ all-MiniLM-L6-v2 (384-dim)
- ▶ FAISS IndexFlatIP
- ▶ LIAR dataset (12,836 claims)
- ▶ Semantic chunking (cosine 0.75)

Deployment & UI

- ▶ Streamlit (app.py)
- ▶ Cloudflare Tunnel (Colab)
- ▶ HuggingFace Spaces
- ▶ pickle / keras .h5 / .safetensors
- ▶ faiss .index + .pkl artifacts



Conclusion & Future Work

Key Achievements

- ✓ Automatic best-model selection at runtime
- ✓ Full RAG pipeline with LIAR fact-check grounding
- ✓ Natural language explanation via FLAN-T5
- ✓ **Hugging face** Streamlit UI with news gate
- ✓ All models serialised and deployment-ready

Future Work

Life Sciences Variant

PUBHEALTH + Monant datasets for medical misinformation

Multimodal Detection

Extend to image + text fake news (deepfake articles)

Real-time Scraping

Plug in live news RSS feeds for continuous monitoring

Azure / AWS Deploy

Scale beyond Colab — containerised cloud deployment

Multilingual

Extend beyond English using multilingual BERT

Feedback Loop

Human-in-the-loop correction for model refinement

User Interface



Fake News Detection System

Paste a news article below and click **Analyze Article**.

The system will:

- Verify if the text resembles a news article
- Predict whether it is REAL or FAKE
- Display confidence score
- Provide a brief explanation

Paste News Article

Analyze Article



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Paste a news article below and click **Analyze Article**.

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Paste News Article

we approach the final capstone presentations

Analyze Article

Input does not appear to be a valid news article.

User Interface

Paste News Article

WASHINGTON (Reuters) - Transgender people will be allowed for the first time in the Trump administration decided not to appeal rulings that blocked his transgender military service. Lower court judges requiring the military to begin accepting transgender recruits. The administration said that it will be releasing an independent study of these issues in the coming months.

Analyze Article

Prediction: REAL

Confidence

99.97%

Explanation

Paste News Article

After Obama was embarrassed by Donald Trump acting more presidential, he decided to find his way off the golf course on Martha's Vineyard to visit the people. The president's course, only Obama could turn the flood into a race issue. While refusing to accept racial discrimination.

Analyze Article

Prediction: FAKE

Confidence

99.86%

Output on Model

Ground truth: 1
Predicted: REAL
Confidence: 0.9997

Top key sentences:

[1] score=0.9997
TOKYO/WASHINGTON (Reuters) - Japan has detected radio signals suggesting North Korea may be preparing another ballistic missile launch, although such signals are not unusual and satellite images did not show fresh activity, a Japanese government source said on Tuesday.

[2] score=0.9988
South Korea s Cho said North Korea may announce the completion of its nuclear programme within a year, as it is moving more faster than expected in developing its arsenal.

[3] score=0.9983
South Korean Unification Minister Cho Myoung-gyon told reporters on Tuesday there have been noteworthy movements from the North since its last missile launch in mid-September, but there was no hard evidence of another nuclear or missile test.

Top LIAR hits:

{'score': 0.466327041387558, 'statement': 'North Koreas missiles are not going to have a capability to reach the United States anytime real soon.', 'label': 'true', 'subject': 'foreign-policy,nuclear', 'speaker': 'john-mccain', 'party_affiliation': 'republican', 'context': 'remarks made on the Fox News segment, Outnumbered'}

{'score': 0.3488151431083679, 'statement': "The nuclear test conducted in our nation this time is the Earth's 2,054th nuclear test. The five permanent members of the U.N. Security Council have conducted 99.99 percent of the total nuclear tests.", 'label': 'true', 'subject': 'military', 'speaker': 'north-korea', 'party_affiliation': 'none', 'context': 'a released statement'}

{'score': 0.3426074683666229, 'statement': 'North Koreahas over two-dozen nuclear warheads.', 'label': 'half-true', 'subject': 'nuclear', 'speaker': 'marco-rubio', 'party_affiliation': 'republican', 'context': 'remarks during a campaign event in Sioux City, Iowa'}

LLM Explanation:

1. Final Verdict 2. Japan has detected radio signals suggesting North Korea may be preparing another ballistic missile launch, although such signals are not unusual and satellite images did not show fresh activity, a Japanese government source said on Tuesday. - South Korean Unification Minister Cho Myoung-gyon told reporters on Tuesday there have been noteworthy movements from the North since its last missile launch in mid-September, but there was no hard evidence of another nuclear or missile test. Retrieved Fact Check Evidence: - The nuclear test conducted in our nation this time is the Earth's 2,054th nuclear test. The five permanent members of the U.N. Security Council have conducted 99.99 percent of the total nuclear tests. (true) - North Koreahas over two-dozen nuclear warheads. (half-true) Similar News 4. Confidence Assessment

Reference Links:

1. Hugging Face link: <https://huggingface.co/spaces/Anirudh765/fakenews-detector/tree/main>
2. Streamlit local host:
3. Data set links:
 - a. <https://www.kaggle.com/datasets/rahulogoel/isot-fake-news-dataset>
 - b. <https://www.kaggle.com/datasets/doanquanvietnamca/liar-dataset>

Thank You

Fake News Detection with Explanation

ISOT + LIAR

BiLSTM + DistilBERT

FAISS RAG

FLAN-T5

Streamlit UI

Questions & Discussion